MedLab Analytics Dashboard

Dashboard Report & Insights Guide

Based on Med_Lab_Cleaned.csv | 2024 Dataset | 155 Records

1. What This Dashboard Is About

The MedLab Analytics Dashboard is a business intelligence tool built in Power BI to help medical laboratory managers and owners understand their operations at a glance. It transforms raw test records from the lab into clear, interactive visuals that answer the key questions every lab needs to monitor regularly.

The dashboard covers five core areas of analysis:

- Revenue — How much money the lab is making and from which services
- Tests — Which lab tests are most popular and in demand
- Patients — Who the lab's patients are and whether they return
- Referrals — Where patients are coming from
- Efficiency — How quickly the lab delivers test results

2. Key Performance Indicators (KPI Cards)

At the top of the dashboard are three KPI cards that provide an instant summary of the lab's overall performance. These numbers update automatically whenever a slicer filter is applied.

KPI Card	Value	What It Means
Total Revenue	**GHS 10.38K**	The total income generated by the lab from all tests performed across the dataset period.
Total Tests	**71**	The total number of lab tests recorded and processed in the dataset.
Avg Turnaround	**4.36 hrs**	The average number of hours it takes from sample collection to result delivery across all test types.

3. Revenue Analysis

3.1 Monthly Revenue Trend (Line Chart)

This line chart shows how the lab's total revenue changes from month to month throughout the year. It helps identify which months are busiest and which are slower, allowing for better staffing and resource planning.

Key Insight:

- Revenue peaks around March, indicating higher patient volume or more expensive tests being ordered during that period.
- A dip in April suggests either fewer patients visited or lower-priced tests were ordered.
- This trend helps the lab prepare for busy seasons and manage quiet periods efficiently.

3.2 Revenue by Test Category (Bar Chart)

This horizontal bar chart ranks all test categories by the revenue they generate. It immediately shows which departments of the lab are the most financially valuable.

Test Category	Revenue (GHS)	Insight
Biochemistry	GHS 6,270	Top revenue category — highest priority for the lab
Hematology	GHS 2,800	Strong second — blood tests are in high demand
Microbiology	GHS 2,470	Significant contributor driven by infection testing
Serology	GHS 2,190	Antibody and disease screening tests
Immunology	GHS 2,020	Immune system related tests
Endocrinology	GHS 1,690	Hormone and thyroid related tests
Cardiology	GHS 1,260	Heart health tests
Virology	GHS 1,200	Virus detection tests including COVID-19

4. Patient Analysis

4.1 New vs Returning Patients (Donut Chart)

This donut chart breaks down the patient base into two groups — new patients visiting for the first time, and returning patients who have been to the lab before.

New Patients	Returning Patients
64.79%	**35.21%**
First-time visitors to the lab	Patients who have visited before

What Retention Means:

Retention refers to how well the lab keeps its existing patients coming back. A retention rate of 35.21% means that out of every 100 patients, about 35 return for a follow-up test. The remaining 65 are new patients who have not returned yet.

- High retention indicates patient satisfaction and trust in the lab
- Low retention may suggest patients are going to competitors for follow-up tests
- Improving retention is more cost-effective than constantly attracting new patients
- The lab should consider loyalty programs or follow-up reminders to improve this rate

4.2 Peak vs Off-Peak Volume (Donut Chart)

This donut chart shows the distribution of tests between busy hours (Peak) and quieter hours (Off-Peak) of the day.

- Peak hours: 47.89% of tests — these are the busiest times, typically morning periods
- Off-Peak hours: 52.11% of tests — quieter periods, usually afternoons
- The fairly even split is a positive sign — it means the lab is managing patient flow well throughout the day
- The lab could consider off-peak discounts to encourage more patients during quiet hours

5. Referral Source Analysis

5.1 Revenue by Referral Source (Bar Chart)

This bar chart shows which sources — hospitals, clinics, or walk-ins — are sending the most patients to the lab and generating the most revenue. This is critical for understanding which partnerships are most valuable.

Referral Source	Revenue (GHS)	Insight
Not Recorded	GHS 7,320	Largest group — referral source not captured at registration
Hospital C	GHS 3,940	Top tracked referral partner — strong hospital relationship

Private Clinic B	GHS 3,930	Nearly equal to Hospital C — valuable clinic partner
Private Clinic D	GHS 3,365	Strong private clinic contributor
Hospital A	GHS 3,305	Consistent hospital referral source
Walk-in	GHS 460	Self-referred patients — small but direct

Important Note on 'Not Recorded':

The largest referral source being 'Not Recorded' (GHS 7,320) is a critical finding. It means the lab is not consistently capturing where patients come from at registration. This should be addressed immediately by training front desk staff to always ask and record referral sources for every patient.

6. Turnaround Time Analysis

6.1 Avg Turnaround by Test Type (Bar Chart)

Turnaround time refers to how long it takes from when a patient provides their sample to when the test results are ready. This chart ranks test types by their average processing time in hours.

What Average Turnaround Means:

- A low turnaround time means faster results and happier patients
- A high turnaround time may indicate the test is complex, requires special equipment, or there is a staffing bottleneck
- The overall average turnaround for the lab is 4.36 hours

Test Type	Avg Hours	Priority to Improve
COVID-19 PCR	8.0 hrs	Highest — needs urgent attention
Allergy Panel	7.5 hrs	Very slow — review process
Thyroid Panel	7.0 hrs	Slow — monitor closely
Vitamin D Test	7.0 hrs	Slow — monitor closely
Cardiac Enzymes	7.0 hrs	Slow — monitor closely
Lipid Profile	6.0 hrs	Moderate — some room to improve
PSA Test	6.0 hrs	Moderate — some room to improve

7. Interactive Slicers (Filters)

The dashboard includes an interactive slicer panel that allows users to filter all visuals simultaneously. This makes the dashboard dynamic — selecting any filter instantly updates every chart on the page.

The slicer covers four filter dimensions:

- Month Name — Filter by specific months to see performance in January, March, etc.
- Test Category — Focus on one department such as Biochemistry or Hematology only
- Referral Source — See performance from a specific hospital or clinic
- Peak/Off-Peak — Compare busy hours versus quiet hours independently

How to Use the Slicer:

- Click any item in the slicer to filter the entire dashboard
- Hold Ctrl and click to select multiple items at once
- Click the eraser icon on the slicer to clear all filters and return to full view

8. Summary of Key Insights

The following are the most important findings from the MedLab Analytics Dashboard that should inform business decisions:

	Biochemistry is the highest revenue-generating category at GHS 6,270 — the lab should invest in growing this department.
	64.79% of patients are new — the lab should focus on retention strategies to convert new patients into loyal returning ones.
	Not Recorded is the largest referral source — the lab must improve how it captures referral data at patient registration.
	COVID-19 PCR has the longest turnaround at 8 hours — this is an area to prioritize for operational improvement.
	Revenue peaks in March — the lab should ensure adequate staffing and resources during this period.
	Peak and Off-Peak tests are nearly balanced (48% vs 52%) — patient flow throughout the day is well managed.
	Hospital C and Private Clinic B are the top tracked referral partners — these relationships should be maintained and strengthened.

9. Recommendations

Based on the dashboard insights, the following actions are recommended for the lab:

Short-Term Actions (Immediate)

- Train front desk staff to always record the referral source for every patient at registration
- Investigate why COVID-19 PCR and Allergy Panel have such long turnaround times and address bottlenecks
- Set up a patient follow-up system — SMS or call reminders — to encourage returning patients

Medium-Term Actions (1–3 Months)

- Develop a loyalty or discount program for returning patients to improve the 35.21% retention rate
- Strengthen partnerships with Hospital C, Private Clinic B and Private Clinic D who are top referral sources
- Consider offering off-peak pricing incentives to balance patient load throughout the day

Long-Term Actions (3–6 Months)

- Invest in equipment or staffing for the Biochemistry department as it is the top revenue driver
- Explore expanding into lower-performing but high-potential categories like Cardiology and Virology
- Publish the dashboard to Power BI Service to allow real-time sharing with management

10. Glossary of Terms

Term	Definition
KPI	Key Performance Indicator — a measurable value that shows how effectively the lab is achieving its objectives
Retention Rate	The percentage of patients who return to the lab for another test after their first visit
Turnaround Time	The time from when a patient provides a sample to when their test result is ready
Peak Hours	The busiest time periods of the day when the lab receives the most patient visits
Off-Peak Hours	Quieter time periods of the day with fewer patient visits

Referral Source	The hospital, clinic, or channel that directed a patient to come to the lab
DAX	Data Analysis Expressions — the formula language used in Power BI to create calculated measures
Slicer	An interactive filter control in Power BI that updates all visuals on the page when clicked
Not Recorded	A referral source value indicating the patient's origin was not captured at registration
Revenue	The total income generated from all test fees paid by patients

End of Report — MedLab Analytics Dashboard 2024